Facial Expression Classification

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Machine Learning for Dynamical
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Homework 1 Report

***Abstract*— In this study, Singular Value Decomposition (SVD), Principal Component Analysis (PCA), and Sparse Representation Classification (SRC) are applied to facial expression images from the FER-2013 dataset. SVD is used to analyze image compression and reconstruction using truncated singular values. PCA is applied to extract eigenfaces and investigate image reconstruction with different numbers of principal components. Finally, SRC is used to classify facial expressions, and the effects of Fourier and Wavelet transformations on classification performance are examined. The results demonstrate the effectiveness of these methods for dimensionality reduction, feature extraction, and facial expression recognition.**

I. Introduction

In this study, facial expression images from the FER-2013 dataset are analyzed using Singular Value Decomposition (SVD), Principal Component Analysis (PCA), and Sparse Representation Classification (SRC) methods.

The main objectives of this homework are:

1.To investigate image compression and reconstruction using Singular Value Decomposition.

2.To extract **Eigenfaces** from facial images using Principal Component Analysis and analyze image reconstruction.

3.To perform facial expression classification using Sparse Representation Classification and evaluate the effect of **Fourier and Wavelet transformations** on classification performance.

All computations in this study were implemented using Python, and the results presented in this report are directly supported by the outputs obtained from the implemented scripts.

II. SINGULAR VALUE DECOMPOSITION

A. Method

Singular Value Decomposition is a matrix factorization method that decomposes a matrix into three matrices as follows:

$$A = U\Sigma V^T, \quad (1)$$

where:

- $U$and $V$are orthonormal matrices
- $\Sigma$is a diagonal matrix containing singular values

For dimensionality reduction, truncated SVD is used:

$$Ar = U_r \Sigma_r V_r^T, \quad (2)$$

where only the first r singular values are retained to reconstruct the image.
To evaluate the quality of reconstruction, the Frobenius norm of the matrix is calculated:

$$\| A \|_F = \sqrt{\sum a_{ij}^2}\,. \quad (3)$$

Additionally, the **compression performance** is evaluated by comparing the original storage size of the image with the storage required by the truncated SVD representation.

B. Results

In this section, a facial image from the dataset is reconstructed using truncated SVD with **r = 5, 20, and 30**.

Figure 1 : Original Angry Face Image

Figure 2 : Truncated SVD with r = 5, 20, and 30.

The Frobenius norms and compression ratios for different values of $r$are summarized in the following table

Table I Frobenius Norm and Compression Ratio for Different Truncation Ranks (r)

r	Frobenius Norm	Compression Ratio
5	7903.777	4.7506
20	7985.827	1.1877
30	7989.035	0.7918

C. Discussion

The results show that the reconstruction quality improves as the value of $r$increases. This occurs because more singular values are preserved, allowing more information from the original image to be retained during reconstruction.

For very small values of r , such as $r = 5$, the reconstructed image appears blurry and only captures the general structure of the face. As the value of $r$increases, the reconstructed image gradually becomes more detailed and closer to the original image.

However, increasing $r$also reduces the compression ratio. This indicates that there is a trade-off between image compression and reconstruction quality.

In this study, the SVD method was applied to a single facial image to examine reconstruction performance. For a more comprehensive analysis, experiments could be conducted on a larger dataset containing multiple facial images. Additionally, images with different lighting conditions, facial orientations, and expression variations could be included to evaluate the generalizability of the method. Furthermore, a wider range of truncation rank values could be tested to analyze the trade-off between compression ratio and reconstruction quality in greater detail. Such experiments would provide a more realistic assessment of the method's performance in practical applications.

III. PCA AND EIGENFACES

A. Method

In this section, **Principal Component Analysis (PCA)** is applied in order to extract the main features of facial images.

Two different facial expressions were selected from the dataset, and **10 images from each expression** were used to form the working dataset.

Each image was vectorized to construct a data matrix. Then the **mean face** was computed as:

$$\mu = \frac{1}{N}\sum xi \,. \quad (4)$$

Next, the dataset was centered by subtracting the mean:

$$X_c = X - \mu \,. \quad (5)$$

Before applying PCA, the data was normalized to ensure **unit variance**:

$$X_{norm} = \frac{X_c}{\sigma} \,. \quad (6)$$

Finally, PCA was performed using Singular Value Decomposition, and the obtained principal components were interpreted as Eigenfaces.

B. Results

1. Mean Face

Figure 3 : Mean Face

2. First Five Eigenfaces

Figure 4 : First Five Eigenfaces

3. Reconstruction Results

A selected test image was reconstructed using different numbers of principal components.

Figure 4 : Reconstruction Results

C. Discussion

Eigenfaces represent the principal variations within the facial image dataset.

When a small number of principal components is used, the reconstructed image only captures the general structure of the face. As the number of principal components increases, more detailed information is preserved and the reconstructed image becomes closer to the original one.

These results demonstrate that PCA is an effective method for **dimensionality reduction and feature representation** in facial image analysis.

In this study, the PCA analysis was performed using a limited number of images. To improve performance and obtain more reliable results, the number of images in the training dataset could be increased. Using a dataset that includes more individuals and facial expressions would allow the computed eigenface components to represent facial variations more effectively. Additionally, the reconstruction error could be systematically evaluated for different numbers of principal components in order to determine the optimal number of components. Moreover, preprocessing techniques such as histogram equalization or image normalization could be applied before PCA to improve the quality of feature extraction.

IV. Sparce Representation classification(SRC)

A. Method

In Sparse Representation Classification, a test image is represented as a linear combination of training images

$$y=Ax, \quad (7)$$

where:

- A is the dictionary matrix containing training images
- x is the sparse coefficient vector

The sparse coefficient vector is obtained by minimizing the L1 norm.

For each class, the reconstruction error is computed as:

$$r_i(y) = ||y - A\delta i(x)||2. \quad (8)$$

The predicted class is determined as the one with the minimum reconstruction error.

B. Results

In this section, the sparse coefficients and residual errors obtained for a test image are presented.

1. Original Image Domain

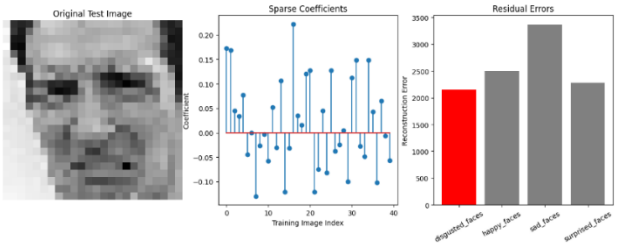

Figure 5 : Test image used for sparse representation classification (original domain).

2. Fourier Domain

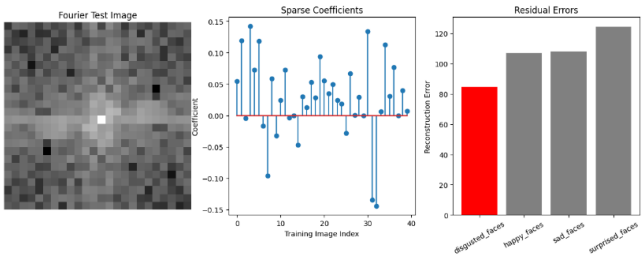

Figure 6: Test image represented in the Fourier domain.

3. Wavelet Domain

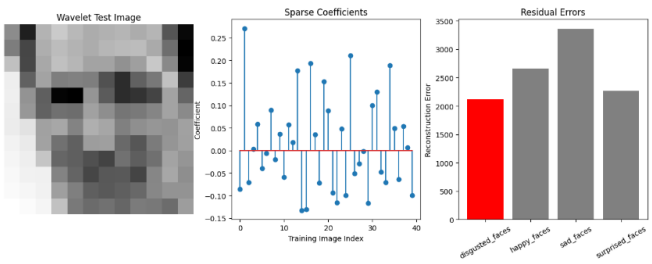

Figure 7. Test image represented in the Wavelet domain.

Table II: Classification Accuracy of SRC in Different Feature Domains

Domain	**Accuracy**
Original	58.33%
Fourier	41.67%
Wavelet	50.00%

C. Discussion

In the sparse representation approach, the test image is represented as a sparse linear combination of the training images. Ideally, the largest coefficients correspond to the images belonging to the correct class.

The results show that the reconstruction error for the correct class is generally smaller than the errors for the other classes, allowing the classifier to correctly identify the facial expression.

When images are transformed into the **Fourier domain**, the representation emphasizes frequency information while partially losing spatial information, which may affect classification performance.

On the other hand, the **Wavelet transform** preserves both spatial and frequency information at multiple scales. Therefore, in some cases it may provide better feature representations and potentially improve classification performance.

The performance of Sparse Representation Classification is highly sensitive to the size and diversity of the training data. Therefore, the classification accuracy could be improved by increasing the number of samples in the training dataset and ensuring a more balanced distribution across expression classes. Additionally, different feature extraction methods, such as PCA-based features or other low-dimensional representations, could be combined with the SRC algorithm to reduce computational cost and improve classification accuracy. Furthermore, testing different regularization parameters and optimization strategies could help obtain more stable sparse coefficients and enhance classification performance.

V. CONCLUSION

In this homework, SVD, PCA, and Sparse Representation Classification methods were applied to facial expression images.

The SVD method provided an effective approach for image compression and reconstruction. PCA successfully extracted eigenfaces representing the main variations within the facial image dataset, and reconstruction quality improved as the number of principal components increased.

Sparse Representation Classification demonstrated the ability to identify facial expressions by representing test images as sparse combinations of training images. Additionally, the effects of Fourier and Wavelet transformations on classification performance were analyzed.

Overall, the results indicate that these methods are effective tools for dimensionality reduction, feature extraction, and classification in facial image analysis.

Although the applied methods produced meaningful results, several improvements could enhance the overall performance of the system. First, increasing the size and diversity of the training dataset could significantly improve the robustness of the classification results. Additionally, applying preprocessing techniques such as image normalization, noise reduction, and contrast enhancement may improve the quality of feature extraction. Future studies may also explore combining dimensionality reduction methods such as PCA with classification algorithms to obtain more discriminative feature representations. Finally, evaluating the models on larger subsets of the FER-2013 dataset and performing cross-validation would provide a more reliable assessment of classification performance.